



Lesson 2.5: Future of Computing

https://codehs.com/lms/assignment/93877835/lesson_plan

Description	Where is computing headed? What is Artificial Intelligence and what are the potential impacts that this might have on our world? In this lesson, students learn about Artificial Intelligence and how the landscape of computing might change in the future. Students will discuss how these future developments might impact our society.						
Objective	Students will be able to: • Discuss the future of technology and computers in the world						
Activities	2.5.1 Video: Future of Computing 2.5.2 Check for Understanding: Video Quiz 2.5.3 Connection: Using DNA for Storage 2.5.4 Free Response: Class Activity: Advancing Technology 2.5.5 Connection: Pros and Cons of AI 2.5.6 Free Response: AI: Is It a Bad Thing?						
Prior Knowledge	• identify important historical events in the development of modern computers						
Planning Notes	• Review the slides and the exercises in the lesson. • This is an excellent lesson to get students talking and discussing important issues. Consider setting up and going over a class code of conduct for successful debates to ensure a safe environment where all voices are heard, valued and considered. • There is a handout that accompanies this lesson. It can be used as an in-class activity or a homework assignment. Determine how and if this handout will be used and make the appropriate number of printouts prior to the class period.						
Standards Addressed	<table><tr><th colspan="2">CSTA 2 Standards</th></tr><tr><th>Name</th><th>Description</th></tr><tr><td>2-IC-20</td><td>Compare tradeoffs associated with computing technologies that affect people's everyday activities and career options.</td></tr></table>	CSTA 2 Standards		Name	Description	2-IC-20	Compare tradeoffs associated with computing technologies that affect people's everyday activities and career options.
CSTA 2 Standards							
Name	Description						
2-IC-20	Compare tradeoffs associated with computing technologies that affect people's everyday activities and career options.						

CSTA 3B Standards

Name	Description
3B-AP-08	Describe how artificial intelligence drives many software and physical systems.
3B-IC-27	Predict how computational innovations that have revolutionized aspects of our culture might evolve.

PRAXIS 5652 Standards

Name	Description
I.A.1	Understand computing as a way of expressing creativity, solving problems, enabling communication, and fostering innovation in a variety of fields and careers
I.A.1.c	provide examples of how computers enable communication and collaboration
I.A.1.d	provide examples of how computers foster innovation
I.A.2	Know the obstacles to equal access to computing among different groups and the impact of those obstacles
I.A.2.c	match obstacles to equal access with effective solutions
I.A.3	Understand beneficial and harmful effects of computing innovations and the trade-offs between them
I.A.3.a	analyze computing innovations in terms of their social, economic, and cultural impacts, both beneficial and harmful
I.A.3.b	identify trade-offs between beneficial and harmful effects of computer innovations

I.B.2	Understand ethical and unethical computing practices and their social, economic, and cultural implications
II.B.1	Be familiar with the limitations of computing in terms of time, space, and solvability as well as with the use of heuristic solutions that can address these limitations
II.B.1.b	recognize the existence of problems that cannot be solved by a computer
II.B.1.c	in context, identify factors that prevent a problem from being solvable
II.B.1.d	identify situations where heuristic solutions are useful
V.A.2	Be familiar with computing systems embedded in everyday objects (e.g., Internet of Things [IoT], ATMs, medical devices)

Teaching and Learning Strategies

Lesson Opener:

- Have students brainstorm and write down answers to the discussion questions listed below. Students can work individually or in groups/pairs. Have them share their responses. [5 mins]

Activities:

- Watch the lesson video and take the corresponding quiz. This quiz is a quick check for understanding. [10 mins]
- Explore the *Using DNA for Storage* connection video. [10 mins]
 - Discuss the pros and cons of this type of technology.
 - What might the students have reservations about regarding computer DNA?
- Complete the *Advancing Technology* class activity. [15 mins]
 - Students can work in pairs or small groups.
 - Consider having each group share one technology example that they brainstormed.
- Explore the *Pros and Cons of AI* connection. [10 mins]
 - Discuss the pros and cons of this type of technology.
 - What might the students have reservations about regarding AI?
- Complete the *AI: Is it a Bad Thing?* free response activity. [5 mins]
- Complete the *The Future is Now* handout. [15 mins]

- This handout can also be used as a homework assignment.

Lesson Closer:

- Have students reflect and discuss their responses to the end of class discussion questions. [5 mins]

Discussion Questions

Beginning of Class:

- Think of something that we use computers for today. How do you think this task was performed in the past before computers? Consider how someone you know who is older than you (an older family member, your teacher, etc) would complete this task.
 - *Sample Response: I use a computer to type my schoolwork. Before computers, I would need to write my notes and assignments by hand.*
- Think of something you or your friends do now that does not use a computer. How can computers help you do it faster or better?
 - *Sample Response: Mow the lawn (although there is tech for that now), order groceries (tech for that too!), drive to school (tech for that - but not widespread), translate for parents (Google translate has existed for a while but AI has recently made leaps in this type of technology), etc.*
- Have you heard of the term “off the grid”? What does this mean and why would people opt to be “off the grid”? Do you know anyone that has gone off the grid?
 - *“Off the grid” is when a person doesn’t have much of a digital presence - no social media, email, etc. People tend to do this to avoid personal information from being stolen or because they prefer face-to-face interactions over virtual ones.*

End of Class:

- In your own words, what is Artificial Intelligence?
 - *AI is the capability of a machine to imitate intelligent human behavior.*
- What potential application of AI excites you the most and why? Share a personal experience or a problem you’ve observed in your community that you think AI could solve.
 - *Answer will vary.*
- Are there any potential applications of AI that concern you? Explain why these applications might be worrying.
 - *Loss of job positions, etc.*

Resources/Handouts

[The Future is Now \(teacher version\)](#)

[The Future is Now \(student version\)](#)

Vocabulary

Term	Definition
------	------------

Computer

A person or device that makes calculations, stores data, and executes instructions according to a program.

Modification: Advanced

Modification: Students Needing Additional Support

Modification: English Language Learners

- Print out video slides for students to reference

- Print out video slides for students to reference